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The Role of Physical Activity in Mild Cognitive Impairment

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Abstract

Introduction: Mild cognitive impairment (MCI) is a common problem in the older population and is associated with an increased risk of developing dementia. Physical activity is one of the important modifiable lifestyle factors that may support cognitive functioning.

Aim: The aim of this paper was to assess the importance of physical activity in people with MCI, with particular emphasis on neurobiological mechanisms, the effectiveness of different forms of training, and their practical application.

Methods: This paper is a narrative review of the literature. Thirty-five publications were included, including original studies, randomized clinical trials, systematic reviews, meta-analyses, as well as guidelines and expert recommendations.

Results: Available data indicate that regular physical activity may have a beneficial effect on memory, executive functions, and global cognitive functioning in people with MCI. Beneficial effects were observed with aerobic training, resistance training, multicomponent programs, and interventions combining physical activity with cognitive training.

Conclusions: Physical activity is an important component of non-pharmacological management in people with MCI. However, due to the heterogeneity of studies and the limited duration of follow-up, there is still a lack of conclusive evidence that it prevents the progression of MCI to dementia.

Keywords: mild cognitive impairment, MCI, physical activity, physical exercise, cognitive functions, older adults

Introduction

Progressive population aging means that cognitive disorders are becoming an increasingly important clinical and social challenge. Mild cognitive impairment (MCI) is of particular importance in the older population and is characterized by cognitive decline greater than expected in the course of physiological aging, with relatively preserved independence in everyday functioning [1,2]. MCI is not equivalent to dementia, but it is associated with an increased risk of

its development; therefore, this period may constitute an important point for undertaking preventive and therapeutic measures [1,2]. In view of the limited possibilities of preventing the progression of cognitive disorders, increasing attention is being paid to modifiable lifestyle factors. One important element is physical activity, the regular performance of which is associated with a beneficial effect on cognitive functioning and may play a role in reducing the risk of developing dementia [3,4,5]. Potential mechanisms of this effect include, among others, improvement in cardiovascular function, support of neuroplasticity, and effects on brain structures involved in memory processes and executive functions [5]. For this reason, physical activity is included as an important component of non-pharmacological management in people with MCI, although the available data do not currently allow a definitive conclusion that it prevents the progression of MCI to dementia [5]. The aim of this paper is to present current knowledge on the role of physical activity in people with mild cognitive impairment, with particular emphasis on the mechanisms of its effects on the brain, the effectiveness of different forms of training, and the possibility of practical use of exercise interventions in the geriatric population.

Methodology

This paper is a narrative review of the literature concerning the effect of physical activity on mild cognitive impairment (MCI). Publications concerning the neurobiological mechanisms of physical activity, the effectiveness of different forms of training, their practical application in older adults, and their effects on cognitive functions were included. The analysis included original studies, randomized clinical trials, systematic reviews, meta-analyses, as well as guidelines and expert recommendations.

The selection of literature focused on issues related to MCI, physical activity, aerobic and resistance training, multicomponent training, cognitive stimulation, and the neurobiological mechanisms of physical activity. Publications from recent years were preferred, although older papers of significant importance to the discussed mechanisms and recommendations were also included. Publications relevant to the aim of the paper were eligible, including both studies directly concerning MCI and papers addressing biological mechanisms and practical aspects of physical activity in older adults. A total of 35 sources were used.

1. Mild Cognitive Impairment (MCI) - Theoretical Foundations

1.1. Definition and Diagnostic Criteria

Mild cognitive impairment (MCI) is a clinical syndrome characterized by a decline in cognitive functions greater than expected for age and level of education, with relatively preserved independence in everyday functioning [1,2]. MCI differs from dementia primarily in the degree to which cognitive deficits affect the patient's functioning - in MCI, basic independence remains preserved, whereas in dementia, cognitive impairment significantly limits the ability to

independently perform everyday activities [1,2]. The classic Petersen criteria included reported memory impairment, preferably confirmed by a close person, objectively demonstrated deterioration of memory relative to age and education, relatively normal general cognitive functioning, preserved ability to perform everyday activities, and absence of dementia [1]. In later concepts, the criteria were expanded to include the possibility of deficits in other cognitive domains, such as executive, language, or visuospatial functions [1,2]. On this basis, amnesic MCI, in which memory impairment predominates, and non-amnesic MCI, which primarily involves deficits in other cognitive functions, are distinguished. Both types may occur in single-domain or multiple-domain forms, depending on the number of affected areas of cognitive functioning [1].

1.2. Epidemiology

Understanding the scale of mild cognitive impairment is extremely important in terms of planning public health strategies and promoting physical activity among older adults. Epidemiological analysis of MCI clearly indicates the growing scale of the problem in aging societies. In the population over 65 years of age, the prevalence of MCI ranges from 15-20%, depending on the diagnostic criteria used and the selected study sample [1,3]. According to the latest reports published in the prestigious *The Lancet Neurology* (2025), MCI is becoming one of the more important challenges for healthcare systems due to global population aging. Moreover, the problem extends beyond the specificity of a single population. A 2025 review showed convergence of conclusions when analyzing Western and Asian countries. Therefore, despite dietary differences and genetic determinants, physical activity remains a universal and the most effective common denominator in MCI prevention [4]. Reports emphasize that without the implementation of effective preventive strategies based on lifestyle modification and risk factors, the number of patients will grow exponentially. Physical activity has been identified as a key component of interventions aimed at improving general fitness as well as protecting the brain's cognitive reserve [6]. With regard to the previously mentioned risk factors that should be taken into account, non-modifiable factors include age and genetic predisposition, particularly the presence of the APOE $\epsilon 4$ allele, while modifiable factors include a sedentary lifestyle, midlife obesity, type 2 diabetes, and hypertension [2]. These are currently at the center of researchers' attention and pose a major threat to cognitive health. Studies show a strong correlation between low cardiorespiratory fitness and poorer results on neuropsychological tests [3]. A diagnosis of MCI is associated with a very high risk of developing dementia, most often Alzheimer's disease. The conversion rate from MCI to fully developed dementia is estimated at 10 - 15% annually in at-risk populations, whereas in healthy populations of the same age it is only 1 - 2% [2]. In summary, the potential to reverse or halt disease progression is a strong argument in favor of the early introduction of exercise rehabilitation, given that according to the 2024 WHO report, more than 31% of the adult population has an insufficient level of physical activity as one of the main modifiable factors, which directly increases the risk of MCI progression.

1.3. Pathophysiology of MCI

The neurobiological mechanisms that are disturbed in the course of MCI demonstrate the importance of the effect of physical activity on brain functioning. An important role is attributed here to hippocampal atrophy, as the hippocampus is important in memory processes, primarily in the consolidation of long-term memory as well as spatial orientation [7]. The latest 2025 review published by Khalil clearly indicated a positive effect of physical activity, including regular walking, in counteracting atrophy of this structure [7]. Volumetric changes are among the most important biomarkers indicating the effectiveness of physical training in patients with MCI, because the hippocampus is a structure that retains the capacity for neurogenesis [7]. Another important aspect is the involvement of vascular factors, since reduced cerebral blood flow - CBF (Cerebral Blood Flow) is characteristic of patients with MCI. Hypoperfusion is observed especially in the frontal and temporal lobes. The temporal lobe is a key area responsible for memory processes, whereas the frontal lobe is responsible for executive functions and cognitive control. Frontal lobe dysfunction is characteristic of non-amnesic forms of MCI. The prefrontal cortex shows high sensitivity to fluctuations in the supply of energy substrates, making it an area particularly dependent on optimal cerebral perfusion. Scientific reports confirm the importance of the effect of physical activity on the cardiovascular system, which directly correlates with better blood supply to brain structures. This phenomenon plays a key role in inhibiting the progression of cognitive deficits, because improved cardiovascular fitness directly translates into the efficiency of metabolic processes in neurons [7,8]. The basic neurodegenerative mechanisms should also not be forgotten, namely pathological processes causing deposition of beta-amyloid plaques and phosphorylation of tau protein. Aggregation of altered tau molecules, destabilization of the cellular skeleton, and disturbances in intraneuronal transport induce neuronal loss through increased oxidative stress, which may be modulated by initiating physical activity [9,10]. Thus, understanding the neurobiological basis of MCI is key to understanding how physical activity may play a therapeutic role. The pathogenesis of this condition is multifactorial and includes neurodegenerative processes, vascular changes, and disturbances of neuroplasticity.

1.4. Diagnosis of MCI

The diagnosis of mild cognitive impairment should be based on a comprehensive clinical assessment, including medical history, assessment of cognitive functions, and the degree of the patient's independence in everyday functioning [1,2]. Both deterioration in cognitive performance reported by the patient or their relatives and objective confirmation of the deficits are important [1,2]. The MoCA (Montreal Cognitive Assessment) and MMSE (Mini-Mental State Examination) tests are primarily used for screening. MoCA is a screening tool sensitive to mild cognitive impairment, whereas MMSE has limited sensitivity in detecting MCI [1]. However, a single screening test result should not constitute the sole basis for diagnosis. In cases of inconclusive results, a more detailed neuropsychological assessment is indicated, including, among others, memory, attention, executive, language, and visuospatial functions [1]. An important element of diagnosis is also the assessment of the ability to perform activities of daily living, because the

relative preservation of functional independence is one of the features distinguishing MCI from dementia [1,2]. Diagnosis should also take into account the patient's somatic and mental status and other possible causes of cognitive deterioration that may influence the clinical picture [1,2].

2.1. Effects of Physical Activity on the Brain - Neurobiological Basis

Neuroplasticity and BDNF

Contemporary neurobiology defines physical activity as a key structural and functional modulator of the central nervous system (CNS). The fundamental adaptive mechanism of the brain to exercise is neuroplasticity, understood as the ability of neurons to create new synaptic connections and form new pathways in response to environmental stimuli. In the context of aging, this process plays an important role by enabling preservation of cognitive reserve despite progressive degenerative changes [11]. At the molecular level, the most important mediator in the process of formation of new neurons induced by physical exercise is brain-derived neurotrophic factor (BDNF). Its increased expression occurs through the functioning of the so-called muscle-brain axis. During skeletal muscle activity, proteins (including irisin and cathepsin B) are released that have the ability to cross the blood-brain barrier and stimulate BDNF production, particularly within the hippocampus [11,12,13,14].

An increase in BDNF concentration leads to a number of beneficial changes in the microstructure of nervous tissue, including:

- (1) enhancement of synaptic transmission by facilitating the phenomenon of long-term potentiation (LTP - Long-Term Potentiation), which constitutes the physiological basis of memory processes [11].
- (2) Neuroprotection: this factor inhibits apoptotic pathways and protects neurons against damage caused by oxidative stress and metabolic disturbances [14].
- (3) Promotion of synaptogenesis: stimulation of dendritic spine growth leads to an increase in the density of connections between neurons, which directly translates into cognitive performance [11,14].

Clinical studies indicate a strong correlation between reduced BDNF levels and the occurrence of dementing diseases. Regular physical activity, through systematic stimulation of neurotrophic pathways, therefore constitutes a proven non-pharmacological strategy in the prevention and treatment of neurodegenerative diseases [12,13].

Angiogenesis and Neurogenesis

In parallel with synaptic changes, physical activity leads to actual remodeling of brain structure. This occurs through two key processes: the formation of new nerve cells (neurogenesis) and the formation of new blood vessels (angiogenesis). For decades, the prevailing belief in science was that the adult brain was incapable of producing new neurons. Studies from the last decade have disproved this myth, demonstrating that this process occurs throughout life, particularly in the dentate gyrus of the hippocampus - an area crucial to our memory. It turns out that regular

movement is one of the strongest stimuli promoting neurogenesis. In practice, this translates into major cognitive benefits: a better ability to remember new events (episodic memory) and more efficient orientation in the environment (spatial memory) [13]. The process of formation of new neurons is inseparably linked to angiogenesis, i.e., expansion of the capillary network. Newly formed nerve cells require immediate access to oxygen and glucose in order to survive. Physical training stimulates overexpression of VEGF (Vascular Endothelial Growth Factor), which initiates the formation of new capillaries [13]. Another important aspect is the effect of exercise on vascular function and so-called brain clearance. Regular physical activity reduces vascular stiffness and improves blood flow. This is crucial for removing toxic metabolic products from the intercellular space, including beta-amyloid deposits and tau protein, as well as for reducing inflammation. In this way, physical activity directly counteracts hypoperfusion and toxin stagnation, which are considered among the primary pathophysiological factors in dementing diseases [15,16].

Metabolic and Molecular Mechanisms

An important mediator between working muscles and the brain is lactate. Lactate is no longer regarded merely as a waste by-product of exercise, but as an important chemical signal that tells the brain to begin building new blood vessels. Lactate activates HCAR1 (Hydroxycarboxylic Acid Receptor 1) receptors, which are present in large quantities on cerebral vascular endothelial cells and pericytes. HCAR1 activation leads to increased production of vascular endothelial growth factor (VEGFA), which directly induces angiogenesis [13]. Increased neuronal network activity during exercise stimulates glutamate receptors, which leads to a massive influx of calcium ions (Ca^{2+}) into the cell. This phenomenon activates calmodulin-dependent kinases (CaMK), which trigger the key transcription factor CREB, promoting expression of the PGC-1 α , NRF1, PPAR α , and TFAM genes as well as the neurotrophic factor BDNF. At the same time, physical exercise induces the activity of mitochondrial sirtuin 3 (SIRT3), which performs a fundamental neuroprotective function in two ways:

1. Through deacetylation of superoxide dismutase 2 (SOD2), it significantly increases its antioxidant potential, protecting the brain against premature aging and cell death.
2. By inhibiting cyclophilin D (CycD), it stabilizes mitochondrial membranes and prevents the opening of pathological pores (PTP), thereby protecting the neuron against cell death. These processes are additionally enhanced by the peripheral fat metabolite 3- β -hydroxybutyrate (3HB). This compound crosses into the brain and stimulates BDNF production in two ways: by strengthening the calcium pathway (Ca^{2+} -CREB) and activating NF- κ B.

Ultimately, binding of the produced BDNF to its receptor (TrkB) activates survival kinase cascades (PI3K, Akt, ERK), which directly translates into increased synaptic plasticity and neuronal resistance to metabolic damage [14].

2.2. Types of Physical Activity Used in Geriatrics

Aging is a complex biological process characterized by a progressive decline in physiological and functional reserve, which significantly increases the risk of frailty syndrome and loss of autonomy.

Key challenges in geriatrics include progressive changes in the musculoskeletal system, including sarcopenia and osteoporosis, as well as reduced cardiovascular capacity and the emergence of cognitive deficits [17]. In light of these changes, physical activity is not viewed solely as a means of improving quality of life, but as an intervention with a broad spectrum of action capable of modifying molecular and cellular processes of aging. Contemporary clinical guidelines, including those published by the World Health Organization [18], clearly indicate that effective exercise programs for older adults must be multidirectional and personalized, focusing not only on individual aspects of fitness but also on their functional integration. For this reason, the scientific literature gives priority to diverse types of exercise, from aerobic exercise through strength training with an emphasis on power to advanced combined training using cognitive stimulation and modern AI technologies, aimed at optimizing effects and minimizing risk.

Aerobic Exercise

Aerobic (endurance) exercise is a foundation of health recommendations, focusing on maintaining cardiorespiratory fitness. At the molecular level, it counteracts vascular endothelial dysfunction by increasing nitric oxide (NO) bioavailability, which reduces arterial stiffness and improves organ perfusion. Of greatest importance in the context of cognitive aging is the effect of aerobic exercise on neuroplasticity. This type of exercise is a strong stimulator of BDNF (Brain-Derived Neurotrophic Factor) expression and secretion in the hippocampus. BDNF supports adult neurogenesis (formation of new neurons in the dentate gyrus) and synaptogenesis, which slows hippocampal atrophy and alleviates symptoms of mild cognitive impairment [19].

Strength and Resistance Training

Strength (resistance) training constitutes the core of non-pharmacological intervention in the aging process, directed specifically against sarcopenia. Contemporary sports geriatrics emphasizes that the key issue is not only maintaining strength, but above all the ability to generate it rapidly, i.e., muscle power.

- **Molecular mechanisms counteracting sarcopenia:** Strength (resistance) training constitutes a strong anabolic stimulus, counteracting age-related protein degradation processes. At the cellular level, mechanical loading initiates activation of the mTORC1 kinase (mammalian Target of Rapamycin Complex 1). This is the main signaling pathway regulating muscle protein synthesis and leading to hypertrophy, as has been extensively described in the physiological literature [20]. Importantly, meta-analyses of clinical studies confirm that this mechanism remains functional even at advanced age. It has been shown that appropriately selected resistance training leads to a statistically significant increase in muscle mass also in very old adults (over 75 years of age), overcoming so-called anabolic resistance [21].
- **Biomechanics and importance of power training:** Muscle power is a better predictor of functional performance and fall risk than maximal strength [22]. Power training counteracts atrophy of type II (fast-twitch) muscle fibers, which are responsible for generating high force and speed over a short period of time. The power-training protocol requires the concentric phase of movement to be

performed at maximum controlled speed, even with moderate loads (40 - 60% of 1 repetition maximum), which optimizes the recruitment of fast motor units.

- Bone health (osteoporosis prevention): Resistance training provides the necessary mechanical load that stimulates mechanotransduction in bone cells. The greatest stimulation of osteogenesis is provided by high loads combined with high dynamics, which increases bone mineral density in the hips and spine [23].

Multicomponent Training

Multicomponent training (MCT) integrates strength, endurance, balance, and coordination for the holistic treatment of frailty syndrome. The effectiveness of MCT results from a synergistic effect on different systems. Improved strength supports movement mechanics, while specific balance exercises target sensory deficits (visual, vestibular, and proprioceptive) necessary for postural control.

The systematic review [17] clearly indicates that programs combining more than three components are the most effective in reducing fall frequency (more than a 30% reduction) and improving global mobility indicators. They prepare older adults for complex everyday tasks.

- Combined training: movement + cognitive stimulation (dual-task training): Dual-task training (DTT) is a response to key functional challenges in older adults arising from the phenomenon of cognitive-motor interference. This phenomenon is defined as an overload of limited attentional resources during simultaneous performance of a motor and cognitive activity, resulting in decreased performance in one or both tasks. In the aging process, physiological deterioration of cognitive and motor functions intensifies this phenomenon, leading to an increased risk of falls in complex everyday situations.

The main goal of DTT is to reduce the so-called "dual-task cost" through systematic training of executive functions, including working memory, cognitive flexibility, and divided attention. At the neurobiological level, this intervention leads to adaptive neuroplasticity involving improved use of neuronal resources in fronto-subcortical circuits. The brain learns to process information more economically, allowing gait to become automated even under conditions of distraction (e.g., during conversation or analysis of the surroundings). Meta-analyses of clinical studies confirm the high effectiveness of DTT. It has been shown that combining motor tasks (e.g., walking, balancing) with cognitive tasks (e.g., counting backwards, verbal fluency) produces better effects in improving gait speed and postural stability than single-component training. Modern technologies such as exergaming, which require rapid decision-making responses to visual stimuli, are also increasingly used for this purpose [24,25].

General WHO and Geriatric Society Recommendations

International guidelines for adults aged 65 years and older provide a framework for clinicians. They emphasize the need for multimodality and individualization of exercise interventions [17,18].

Recommendations Concerning Duration, Intensity, and Type of Activity

- **Aerobic activity:** According to WHO recommendations, 150 to 300 minutes of moderate-intensity exercise or 75-150 minutes of high-intensity exercise per week is recommended [18]. In geriatrics, it is recommended that this training be performed at least 3 times per week in sessions lasting 30 - 45 minutes [17].
- **Strengthening training:** Strengthening of the major muscle groups at least 2 times per week, with a load of 70 - 80% of 1 maximal load. Muscle power training is the priority [22].
- **Balance training:** 2 or more times per week, exercises improving balance and functional strength [17].
- **Limiting sitting:** It is recommended to minimize time spent in a seated position by replacing inactivity with movement.

Limitations and Precautions

- **Medical qualification:** Before beginning a higher-intensity program, medical consultation is required, especially in the presence of unstable coronary artery disease, uncontrolled hypertension, or advanced osteoporosis.
- **Initial assessment:** Functional tests (SPPB - Short Physical Performance Battery, TUG - Timed Up and Go) and frailty assessment scales must be used in order to adjust intensity.
- **Technique safety:** The Valsalva maneuver should be avoided during strength exercises, as it rapidly increases intrathoracic and intracranial pressure, increasing the risk of vascular events.
- **Environmental adjustment:** Elimination of fall hazards and monitoring of hydration and thermoregulation are crucial [17].

3. Scientific Evidence on the Effect of Physical Activity on MCI

The risk of developing mild cognitive impairment and dementia over a 10-year perspective and over the course of a lifetime increases with increasing biological severity of Alzheimer's disease in people who are currently cognitively unimpaired. The biological severity of Alzheimer's disease was assessed on the basis of centiloid values in amyloid PET imaging, and an increase in this indicator was associated with an increased risk of developing cognitive impairment [6]. Therefore, preventive strategies based on modifiable lifestyle factors, including physical activity, are being sought [5,26].

3.1. Aerobic Exercise and Cognitive Functions

The main focus of research has been on aerobic exercise, which is the most commonly described intervention in the literature [26]. It has been shown that regular physical activity, including aerobic training, may lead to improvement in cognitive functions, particularly executive functions and memory, in people with mild cognitive impairment [26]. Guidelines concerning MCI indicate that training conducted for approximately 6 months probably improves cognitive-function measure

outcomes in this group of patients [2]. Physical activity may also be associated with structural changes within the brain. A review of studies on regular walking demonstrated beneficial changes in the volume of hippocampal formation structures, although the magnitude of the effect varied depending on the type and intensity of the intervention [7].

3.2. Strength and Resistance Training

Although resistance training has so far been analyzed less frequently than aerobic exercise, available results indicate its beneficial effect on cognitive functions in people with MCI, particularly memory and executive functions [26,27]. In a study involving 24 weeks of resistance training in older adults with MCI, improvement in verbal episodic memory and beneficial changes in white matter integrity were demonstrated. A protective effect against reduction in the volume of the right hippocampus and right precuneus was also observed in the training group [28]. A 2026 network meta-analysis further indicated that, among the types of activity analyzed, resistance training ranked highest in terms of its effect on executive functions [27].

3.3. Multicomponent Programs

Multicomponent programs most often integrate aerobic and resistance exercise as well as elements improving balance and coordination. In a meta-analysis including 55 randomized clinical trials and 4,603 people with MCI, all analyzed forms of exercise had a beneficial effect on global cognitive functions compared with a passive control group, while multicomponent programs ranked highest in the effectiveness ranking [27]. Such an approach allows simultaneous effects on different components of physical and cognitive performance. However, the considerable diversity of the protocols used in terms of type of exercise, frequency, intensity, and duration continues to make it difficult to determine a single optimal model of intervention [5,27].

3.4. Physical Activity Combined with Cognitive Training

Interventions combining physical activity with cognitive stimulation are attracting increasing interest. A 2024 meta-analysis showed that such programs may lead to improvements in global cognitive functions, memory, and executive functions in older adults with MCI [29]. With regard to executive functions, combined interventions were superior to physical training alone, whereas no clear superiority over cognitive training alone was found [29]. In a randomized clinical trial involving 95 older people with MCI, a 12-week program combining physical activity with cognitive training led to improvement in both physical performance and cognitive outcomes, including executive functions and verbal fluency [30]. These results indicate the potential of combined interventions; however, further studies directly comparing them with single forms of therapy are needed.

3.5. Importance of Intervention Duration

The duration of an exercise program may be important for the cognitive effects achieved. In the most recent meta-analysis, the largest effect estimates for multicomponent programs were observed with interventions lasting 13–24 weeks, although duration itself was not a statistically

significant moderator of the improvement obtained [27]. Available guidelines also indicate probable cognitive benefits resulting from training programs conducted for approximately 6 months [2]. Data concerning maintenance of effects after completion of the intervention and the long-term course of MCI remain limited. Reviews of long-term studies suggest the possibility that some cognitive benefits may persist; however, the current quality of evidence does not allow definitive conclusions to be drawn [31]. It has also not yet been conclusively demonstrated that regular physical activity prevents the progression of MCI to dementia [5,31].

4. Practical Aspects of Incorporating Physical Activity into MCI Therapy

4.1. Selection of Exercise for Geriatric Patients

Personalization of the Intervention:

The effectiveness of therapy in geriatric patients requires certain conditions to be met. One of these is personalization of the intervention for the geriatric patient. The group of geriatric patients is highly heterogeneous in many respects; although they belong to one age group, they have different functional statuses, different levels of fitness, and may have chronic diseases. A "one-size-fits-all" approach is insufficient here [32]. Personalization of the intervention is therefore necessary.

One of the foundations of personalization is adjustment of the exercise level to functional fitness individually for each patient. Attention should also be paid to the frequently present frailty syndrome. Training strategies and exercise plans are divided into levels of advancement. They range from simple functional exercises to complex ones. Each strategy is based on the individual patient's level of fitness [5]. Consideration of the patient's clinical condition and tolerance of physical exertion is necessary when selecting exercise intensity, frequency, or load [33]. Practically every increase in exercise load should be correlated with the individual response of each patient's body; as a result, long-term therapy becomes more effective and complications are minimized.

An important aspect influencing exercise selection and its modification is the presence of comorbidities and their severity. Therapeutic modifications are determined depending on the presence of vascular, musculoskeletal, or neurological disorders [17]. Today, dynamic personalization is a key factor in improving patients' physical functions and quality of life. Exercise programs are continuously adjusted to the patient's needs and capabilities during therapy. Regular assessment of progress and modifications of training plans significantly improve the effectiveness of therapy [26].

In practice, technologies supporting personalization of interventions may be used. Physical-activity monitoring systems or programs based on digital tools are available. Use of the available tools is helpful in adjusting individual exercise programs to the patient's current level and monitoring progress [34].

In summary, personalization of interventions in geriatric patients allows a significant increase in intervention effectiveness and improves safety during physical activities. This process constitutes

a multifactorial approach based on individual therapeutic goals, exercise tolerance, comorbidities, and possibilities for progression.

Assessment of Functional Status and Fall Risk:

The level of physical fitness strongly correlates with the frequency of falls among older adults; early studies showed that regular assessment of functioning is necessary in this age group [32]. An important factor influencing the relationship between functional status and fall risk is frailty syndrome. It has been demonstrated that frailty is strongly associated with reduced physiological reserve and adaptive capacity of the body. These factors have a negative impact because they increase susceptibility to injuries and adverse events [5].

Functional assessment of the geriatric patient should be expanded to include an in-depth analysis of factors affecting postural stability and the ability to control performed movements. Components of physical fitness such as muscle strength, motor coordination, locomotor abilities, and balance are major elements in the assessment of the overall physical fitness of the geriatric patient. Studies show a strong correlation between reduced muscle strength, particularly in the lower limbs, and balance disorders and the risk of falls in this population [33].

Comorbidities are another important factor influencing assessment of the patient's functional status and fall risk. Cardiovascular, neurological, and musculoskeletal diseases strongly increase the risk of falls in geriatric patients through balance disturbances or limited mobility [17]. Comprehensive functional assessment of the patient should include both physical and clinical aspects individualized to each patient in this population.

Therapeutic strategies require regular updating. The functional status of older adults may deteriorate rapidly due to comorbidities, repeated hospitalizations, or limitations related to physical activity [26]. By monitoring the patient's general condition and regularly adapting therapeutic needs to the exercise program, it is possible to detect deterioration in physical performance early and implement preventive measures.

In summary, analysis of physical performance and regular updating of health status and patient capabilities allow optimization of the intervention, improvement of quality of life, and reduction in the risk of failure or injury.

4.2. Training Safety

Contraindications:

Training safety is another very important aspect in planning a therapeutic strategy for a geriatric patient. Physical activity in this population produces visible therapeutic benefits; however, its implementation should be preceded by an in-depth risk analysis and early identification of contraindications [32].

A characteristic group of patients are people with frailty syndrome. Due to the significantly increased risk of injury, particular caution is required when selecting exercises [5]. This group of patients has an increased risk of falls, limited exercise tolerance, or existing balance disorders.

Training safety criteria are based on precisely defined exercise parameters and principles of monitoring. Early detection of symptoms of exercise intolerance is essential in order to maintain safety during exercise; individual adjustment of exercise intensity to each patient is necessary [33]. Contraindications that we define as relative include, among others, severe dyspnea, dizziness, and chest pain. All of these should be a signal to stop training or reduce its intensity. Contemporary methods also draw attention to the safety of training in home conditions. Appropriate environmental conditions and the absence of supervision are worth mentioning here [34]. Neither of these two components is a classic contraindication to training, but they are important aspects that should be considered when planning a training strategy or individually modifying the training plan for each patient. Eliminating potential hazards and ensuring optimal conditions results in increased effectiveness of therapeutic measures.

Health Status Monitoring:

Monitoring health status in this population is an integral element of every stage of the intervention. The implementation of physical activity should always be associated with regular assessment of the geriatric patient's response to exercise; this ensures effective detection of adverse changes and, consequently, appropriate and rapid modification of the exercise program [32].

Patients with frailty syndrome are particularly vulnerable to training overload. In this group, even small changes in training load may result in overload. Continuous monitoring of functional parameters and subjective symptoms is necessary [5].

In clinical practice, both scales of physical-exertion assessment based on subjective evaluation and objective physiological indicators are used. Multidimensional assessment allows early detection of potential deterioration and its prevention. The literature indicates that particular caution is required when symptoms such as dyspnea, fatigue, dizziness, or pain occur. They may be the first sign of significant overload of the body [33].

Comorbidities are another important element influencing the effectiveness of the therapy applied. Health monitoring also consists of continuous observation of parameters associated with the functioning of systems such as the cardiovascular, respiratory, and neuromuscular systems [17]. Observation of these parameters makes it possible to detect deterioration in the patient's general condition early and prevent complications associated with physical activity.

Monitoring health status in this population is an integral element of therapy. It is multidimensional in nature. It forms the basis for maintaining the safety of therapy and its long-term effectiveness.

Recommendations Concerning the Intensity of Physical Exercise

The strongest scientific evidence concerns activity of moderate or moderate-to-vigorous intensity (MVPA). Moderate exercise corresponds to 50–70% of maximum heart rate (HR_{max}) or 3 - 6 MET (Metabolic Equivalent of Task), where 1 MET corresponds to resting energy expenditure. High-intensity training exceeds 70% HR_{max} (>6 MET) [8,32]. Studies indicate that this range of

intensity is the most effective for improving cognitive functions, particularly episodic memory and executive functions [17,34].

According to recommendations based on analyses of clinical interventions, the optimal "dose" of physical activity is at least 150 minutes per week of moderate exercise or, alternatively, 75 minutes of vigorous exercise, distributed across several training sessions [3,5,32]. The most commonly recommended exercise is distributed across 3- 5 training sessions per week, lasting 30 to 60 minutes. Importantly, a dose-response relationship has also been demonstrated, indicating that a greater volume of activity may be associated with a stronger neuroprotective effect [26].

In terms of type of activity, the best-documented benefits concern aerobic training, which improves cerebral perfusion and stimulates neurogenesis within the hippocampus [7]. However, increasing evidence indicates that multicomponent programs combining aerobic and resistance training with balance and coordination exercises are the most effective [5,17]. Strength training additionally supports executive functions, probably through hormonal and metabolic mechanisms [8].

Mechanisms underlying the beneficial effect of physical activity include, among others, increased expression of neurotrophins (particularly BDNF), reduction of inflammation, improvement in endothelial function, and an increase in the volume of brain structures crucial for memory, such as the hippocampus [6,7,24]. Systematic reviews have shown that regular walking and aerobic activity may lead to measurable changes in hippocampal volume, which directly correlate with improvement in cognitive functions [7].

4.3. Models for Implementing Physical Activity in Practice

Despite well-documented benefits, the effectiveness of interventions depends to a large extent on the way they are implemented. Three main implementation models are distinguished in the literature: clinical programs, community-based interventions, and telemedicine-based solutions.

Clinical Programs

Programs delivered in clinical settings constitute the most controlled and best-studied form of intervention. They usually include individually tailored training plans carried out under the supervision of a team of specialists, including physicians, physiotherapists, and neuropsychologists [5,32].

These interventions often combine physical activity with cognitive training, which allows a synergistic effect to be achieved. Studies indicate that such an approach may lead to significant improvement in cognitive functions and slowing of MCI progression [1,17]. These programs usually last from several months to a year and include regular moderate-intensity training sessions.

The main limitation of this model, however, is high cost and limited availability, which hinders its widespread population-level use [5].

Community-Based Programs

An alternative to clinical interventions is programs implemented in the local community, such as classes in senior clubs, walking groups, or Nordic walking training. Such activities are characterized by high accessibility and low cost, making them particularly attractive from a public-health perspective [3,8].

An important element of these programs is the social component, which promotes increased motivation and reduces social isolation, a risk factor for deterioration of cognitive functions. Regular participation in group forms of activity is associated with improvement in global cognitive functions and psychological well-being [3].

Studies have shown that even simple forms of activity performed in the company of other people, such as regular moderate-intensity walks, may significantly reduce the risk of developing dementia, particularly when undertaken over the long term [26].

Telemedicine and Digital Technologies

The dynamic development of digital technologies has enabled the implementation of telemedicine-based interventions, including mobile applications and wearable devices. These solutions allow monitoring of physical-activity levels, control of physiological parameters, and delivery of personalized recommendations [24].

Studies indicate that the use of mobile applications may increase adherence to physical-activity recommendations, which is a key factor in intervention effectiveness. In addition, the possibility of remote patient monitoring enables rapid modification of the training program. Despite numerous advantages, implementation of such solutions is associated with limitations such as digital exclusion of older adults and variable quality of the collected data [24].

5. Limitations of Current Research and Directions for Future Studies

5.1. Heterogeneity of Interventions

Analysis of the available studies indicates high heterogeneity of interventions used in the population of people with mild cognitive impairment (MCI) [5,27]. In addition to traditional exercise programs including aerobic and resistance training, more complex forms also appear, such as therapeutic tango integrating movement, music, and social-cognitive elements, mind-body exercises, and multicomponent programs, which significantly complicates comparison of results and formulation of unequivocal conclusions [27,35].

Important differences also concern the intensity, frequency, and duration of individual training programs [5,27]. Moreover, different types of physical activity may have different effects on individual cognitive domains, further making it difficult to identify a single optimal model of intervention [27].

5.2. Methodological Differences

An important limitation of the analyzed literature is clear methodological inconsistency and problems related to participant selection [5,27,31]. Different tests and tools for assessing cognitive functions were used in individual studies, which limits the possibility of directly comparing the obtained results [27,31]. Differences in the characteristics of the study populations, including sex, level of education, and the setting in which the intervention was conducted, may further limit the generalizability of the results [31]. An additional limitation is the risk of bias, which was not the same in all analyzed studies [27]. Implementation of exercise protocols may also be complicated by coexisting frailty and limitations in physical performance occurring in part of the geriatric population [33].

5.3. Short Follow-up Duration in Most Studies

Another important limitation is the relatively short duration of many interventions and of follow-up after their completion. A substantial proportion of training programs in people with MCI last from several weeks to several months, which allows assessment of short-term changes in cognitive functions but limits the possibility of drawing conclusions about the durability of the effects obtained [27,31].

Data on the long-term maintenance of benefits after completion of regular training remain limited. Individual findings suggest the possibility of a persistent beneficial effect on global cognitive functioning; however, results concerning other measures remain inconsistent and do not allow unequivocal conclusions to be drawn [31].

5.4. Lack of Conclusive Data on Preventing the Transition from MCI to Dementia

Despite numerous reports indicating a beneficial effect of physical activity on selected cognitive functions in people with MCI, the available evidence regarding prevention of MCI progression to dementia remains inconclusive and insufficient [5,31]. One important problem is the lack of consistency in the tools used to assess cognitive functions, which makes comparison of results and formulation of unequivocal conclusions difficult [27,31].

In many studies, the follow-up period is also insufficient to assess the effect of an intervention on the development of dementia, which is a long-term process [31]. Data concerning the association between physical activity undertaken at earlier stages of life and later risk of dementia come primarily from observational studies, which limits the possibility of inferring a causal relationship [3]. Therefore, improvement in cognitive-test results observed in some studies should not be equated with proven prevention of conversion from MCI to dementia [5,31].

5.5. Proposed Directions for Future Research

In order to obtain more conclusive data on the effect of physical activity on the course of MCI, future studies should include larger groups of patients and longer follow-up periods [27,31]. It is also important to standardize intervention protocols by precisely defining the type of exercise, its

intensity, frequency, and duration, taking into account comorbidities, baseline fitness, and the patient's capabilities [27,31].

The feasibility and costs of proposed interventions as well as the potential of simple and easily accessible forms of physical activity, such as walking, should also be taken into account [7,26,31].

Another promising direction for future research is the use of neuroimaging methods and analysis of biomarkers and mediators such as BDNF, which may allow a better understanding of the mechanisms by which physical activity affects specific brain structures and functions [7,12].

Discussion

Available data indicate that physical activity may have a beneficial effect on cognitive functions in individuals with MCI, particularly memory and executive functions [5,27]. Beneficial outcomes have also been observed with multicomponent programs and interventions combining physical exercise with cognitive training [27,29].

However, the variability of the observed effects suggests that the benefits obtained may differ depending on the type of training used and intervention parameters such as intensity, frequency, and duration [27]. Available findings suggest that different forms of exercise may affect different cognitive domains, which makes it difficult to identify a single optimal intervention model for all patients with MCI [5,27]. Therefore, individualization of the physical activity program, taking into account the patient's functional status, comorbidities, and capabilities, is of particular importance [27].

Interpretation of the results is limited by methodological differences, heterogeneity of interventions, and the short duration of follow-up [5,27,31]. In many studies, primarily short-term changes in cognitive test scores were assessed, whereas data on the durability of the observed effects remain limited [31]. Improvements in cognitive functions observed during interventions should not be equated with proven slowing of disease progression. There is still a lack of conclusive evidence that physical activity prevents the progression of MCI to dementia [5,31].

Conclusion

Physical activity is an important component of non-pharmacological management in people with MCI and may support cognitive functions and physical performance [5,27,29]. Regularity, individualization, and appropriate selection of the type of training are important [5,27]. Current data are promising; however, further studies are needed regarding long-term effects and the influence of physical activity on the progression of MCI to dementia [5,31].

Disclosure

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